
Group 04

**SmartSpend AI
Vision Document**

Version 1.0

SmartSpend AI	Version: 1.0
Vision Document	Date: 10/07/26
SS-VSD-01	

Revision History

Date	Version	Description	Author
10/07/26	1.0	Initial draft for SmartSpend AI Vision Document	Tạ Bình Phong Nguyễn Trần Lan Viên

SmartSpend AI	Version: 1.0
Vision Document	Date: 10/07/26
SS-VSD-01	

Table of Contents

1. Introduction	4
2. Positioning	4
2.1 Problem Statement	4
2.2 Product Position Statement	4
3. Stakeholder and User Descriptions	5
3.1 Stakeholder Summary	5
3.2 User Summary	5
3.3 User Environment	5
3.4 Alternatives and Competition	6
4. Product Features	6
5. Non-Functional Requirements	8
5.1 Efficiency	8
5.2 Availability & Reliability	9
5.3 Security Requirements	9
5.4 Compatibility & Portability	9

SmartSpend AI	Version: 1.0
Vision Document	Date: 10/07/26
SS-VSD-01	

Vision (Small Project)

1. Introduction

The purpose of this document is to collect, analyze, and define the high-level needs and features of the **SmartSpend AI** system.

SmartSpend AI is a personal expense management mobile application designed to help users track their daily spending, manage monthly budgets, and better understand their financial habits. This document focuses on the capabilities needed by stakeholders and target users, and the reasons these needs exist. The details of how SmartSpend AI fulfills these needs will be specified in the Use-Case Model and Software Requirements Specification (SRS).

2. Positioning

2.1 Problem Statement

The problem of	Difficulties in personal financial management, frequently forgetting to record small expenses, time-consuming manual data entry, improper expense categorization, and a lack of clarity regarding monthly cash flow.
affects	Young users, especially students and early-career employees (ages 18 - 27).
the impact of which is	Unexplained financial losses, exceeding budgets, and a high abandonment rate of expense tracking apps due to the excessive time required for manual input.
a successful solution would be	An application that minimizes recording effort through AI technology (receipt scanning, auto-categorization), provides clear visual reports, and issues timely budget warnings.

2.2 Product Position Statement

For	Students and young people with low income ranging from 2 million to 20 million VND
Who	Need a simple and convenient tool to track expenses, set budgets, and receive spending insights.
The (product name)	SmartSpend AI
That	Automates expense recording via OCR, automatically categorizes expenses and provides smart budget warnings.
Unlike	Traditional manual tracking methods (notebooks, sheets), basic expense tracking apps.

SmartSpend AI	Version: 1.0
Vision Document	Date: 10/07/26
SS-VSD-01	

Our product	Maximizes time-saving in data entry with AI, delivering a fast and convenient experience to ensure users do not abandon their financial tracking habits.
-------------	--

3. Stakeholder and User Descriptions

3.1 Stakeholder Summary

Name	Description	Responsibilities
Supervisor	Teaching Assistant overseeing the project.	- Provide their guidance throughout the development process of the SmartSpend AI application. - Review website documents, design and source code.
Development team	Development team responsible for the platform development	- Prepare the technical documentation for software development. - In charge of developing and constructing the platform with fully functional features. - Ensure the quality of the product.
Platform operator	Operate and manage the platform.	- Manage platform users.
IT Department	Technical team responsible for system maintenance	- Ensure the system is secure. - Provide technical support. - Implement system updates.
AI Service Provider	Third-party (or internal) AI system/service.	Receives receipt images from the app, performs OCR to extract data, predicts categories, and returns the results to the system.

3.2 User Summary

Name	Description	Responsibilities	Stakeholder
Administrator	System administrators with full access privileges	- Manage user permissions - Oversees system operations - Monitors system performance - Handles critical issues	IT Department Platform operator
User	Includes students and office workers needing expense management.	- Logs in. - Records transactions(manually or via AI scanner). - Manages budgets. - Views financial reports. - Updates personal profile.	

3.3 User Environment

Administrator:

SmartSpend AI	Version: 1.0
Vision Document	Date: 10/07/26
SS-VSD-01	

- Number of people: 1-3 system administrator
- Task cycle: Review system logs and user feedback weekly (30-60 minutes), handle emergency troubleshooting when necessary.
- Environment: Work primarily on desktop computers in an office setting with a stable internet connection.
- Platforms: Web-based Admin Dashboard using modern browsers (Chrome, Edge).

Customer (End-User):

- Number of people: Individual users managing their own private financial data.
- Task cycle: High-frequency, short-duration tasks. Adding a transaction via AI scan takes 10-15 seconds. Reviewing monthly charts takes 2-5 minutes.
- Environment: Mobile. Users will interact with the app on the go—at checkout counters, cafes, or on public transport right after a purchase.
- Platforms: Android and IOS smartphones.
- Integration needs: Interacts indirectly with mobile banking apps or e-wallets (users take screenshots of transfer receipts from these apps and upload them to SmartSpend AI).

3.4 Alternatives and Competition

Methods and tools used in personal finance management within Vietnam are identified as SmartSpend AI's competitors. They include: Physical Notebooks/Excel, traditional budgeting apps (Money Lover, Spendee), and digital e-wallets (Momo, ZaloPay). Through discussion with the development team, key characteristics of each platform is summarized as following:

- Physical Notebooks / Excel Spreadsheets: This traditional method is completely zero cost and guarantees 100% absolute data privacy. However, it requires highly tedious manual entries, users easily forget to record minor daily expenditures, and it lacks automated category tagging or proactive budget limit alerts.
- Traditional Budgeting Applications: Platforms like Money+ or Spendee showcase well-optimized user interfaces and excellent graphical charts. Nevertheless, their free tiers rely heavily on repetitive typing which drives high drop-out rates, and smart features like receipt scanning require paid premium upgrades and lack optimization for Vietnamese character parsing.
- E-Banking & E-Wallet Analytics: Applications such as Momo or Mobile Banking apps offer automated electronic transaction logging with bank-grade security systems. However, the financial data remains fragmented across separate apps, it completely fails to log physical cash transactions, and it does not support custom sub-category limits.

Through analysing competitors' flaws, our platform aims to deliver an optimised, mobile-first experience, specially tailored to Vietnamese students and young adults. Our platform will combine the strengths of our competitors while addressing their weaknesses through smart AI automation.

4. Product Features

This section describes and defines high-level features of the SmartSpend AI application. Features are the high-level capabilities of the application that are necessary to deliver benefits to the users.

No.	Feature	Description	Priority
1	Account Registration via Email	The application shall allow users to register a new account via Email.	High

SmartSpend AI	Version: 1.0
Vision Document	Date: 10/07/26
SS-VSD-01	

2	Password Authentication	The application shall allow users to log in to the system using a traditional password.	High
3	Third-Party Google Login	The application shall allow users to log in quickly via a third-party account (Google).	High
4	Password Recovery System	The application shall allow users to recover their password by sending an OTP code or a password reset link via their registered Email.	High
5	Profile Information Display	The application shall display the user's personal information, including avatar, full name, email, and join date.	High
6	Profile & Password Update	The application shall allow users to update their personal information and change their password.	High
7	Custom Category Creation	The application shall allow users to create new custom expense categories (e.g., Dining, Education, Transportation).	High
8	Custom Category Management	The application shall allow users to view, edit, or delete their custom-created categories.	High
9	Manual Transaction Logging	The application shall allow users to add a new manual transaction by inputting the amount, date and time, notes, and corresponding category.	High
10	Timeline History Display	The application shall display the transaction history in a timeline format.	High
11	Advanced Transaction Filter	The application shall provide an advanced filter allowing users to search for transactions by date, amount range, category, or specific keywords.	High
12	Transaction Modify & Delete	The application shall allow users to edit information or delete previously created transactions.	High
13	Monthly Budget & Savings Setup	The application shall allow users to set a total monthly budget (intended spending amount) and a savings goal.	High
14	Category Spending Ceilings	The application shall allow users to set a maximum spending limit for specific categories (e.g., a maximum of 2 million/month for Shopping).	High
15	Visual Budget Progress Tracking	The application shall display a visual progress bar indicating the percentage of the budget used	High
16	Over-limit Red Alerts	The application shall automatically display a warning notification (in red) when a user's spending reaches 80% to 100% of the established limit.	High

SmartSpend AI	Version: 1.0
Vision Document	Date: 10/07/26
SS-VSD-01	

17	Historical Budget Review	The application shall allow users to review the budget history of previous months to evaluate goal completion.	High
18	Transaction Image Upload	The application shall provide an interface allowing users to upload or directly capture images of receipts and bank transfer screenshots.	High
19	Automated OCR Extraction	The application shall use OCR technology to automatically extract information from receipt images (Amount, Store, Date and Time) and pre-fill the form, requiring user confirmation before saving.	High
20	AI Auto-Categorization	The application shall use AI to automatically analyze the store name from the receipt and suggest or assign an appropriate expense category.	High
21	Categorical Expenditure Chart	The application shall display a Pie Chart illustrating the percentage breakdown of spending by category for the current month.	Medium
22	Financial Trend Comparison	The application shall display a Bar/Line Chart comparing the trend of total income versus total expenses across different months.	Medium
23	Financial Data Export	The application shall allow users to export financial history data to Excel or CSV formats for storage.	Medium
24	Application Configurations	The application shall allow users to configure application settings, including toggling Dark mode, selecting the language, and choosing the currency (VND/USD).	Medium
25	Recurring Entry Reminders	The application shall allow users to schedule recurring reminder notifications for data entry (e.g., at 9:00 PM daily).	Medium
26	Summary Main Dashboard	The application shall provide a Dashboard on the main screen summarizing the current balance, summary charts, and quick-action buttons.	Medium
27	System Admin Statistics	The application shall provide an administrative interface specifically for Admins to view overall system statistics (e.g., total number of users).	Low
28	Bug & Feedback Management	The application shall allow Admins to view and manage feedback, suggestions, or bug reports submitted by users.	Low

5. Non-Functional Requirements

5.1 Efficiency

- Response time:
 - + The system must respond to standard user actions (e.g., login, manual entry logging, history filtering, chart rendering) within 1.5\$ seconds in 95% of cases.

SmartSpend AI	Version: 1.0
Vision Document	Date: 10/07/26
SS-VSD-01	

- + The cumulative execution of the smart AI pipeline (receipt/screenshot OCR scanning and auto-categorization) must resolve within 4.0 seconds.
- Load Handling: The platform should maintain absolute stability and responsiveness during high-traffic hours when concurrent users are logging expenditures (e.g., end-of-day periods or holiday seasons).

5.2 Availability & Reliability

- Fault Tolerance: In case of system errors or third-party API connectivity losses, users should receive meaningful, user-friendly error messages, and the application should automatically redirect to the manual input form to recover gracefully without crashing.
- Auto Recovery: The system must be able to restart critical backend services and AI classification pipelines automatically if they fail or disconnect.
- Offline Operational Reliability: The mobile client must support offline manual entry logging utilizing local data caching, with automated synchronization back to the cloud database once internet connectivity is re-established.

5.3 Security Requirements

- Authentication & Authorization: Implement role-based access control (RBAC) to dynamically isolate individual consumer profile boundaries and securely unlock the hidden administrative features for the Single Admin Account upon authentication.
- Data Protection:
 - + Passwords must be stored in the database using strong, secure one-way hashing algorithms.
 - + Sensitive financial records and personal data (emails, account balances, transactional history) must be encrypted in transit via HTTPS and at rest within the database system.

5.4 Compatibility & Portability

- Responsive Mobile Layout: The mobile application interface must be fully responsive and optimized for different smartphone screen sizes and resolutions to ensure visual clarity.
- Platform Compatibility: The front-end user client along with the embedded administrative control panel must deploy natively and function correctly on the Android mobile operating system (iOS if possible).
- Data Portability: The platform must support data portability by enabling users to seamlessly export their historical transactional data from the database into standard Excel or CSV spreadsheet formats.
- Third-Party Service Compatibility: Any third-party services, utility libraries, or smart cloud APIs (OCR and LLM models) utilized within the application must fully support the target Android ecosystem.